PulseView – Quick start Guide

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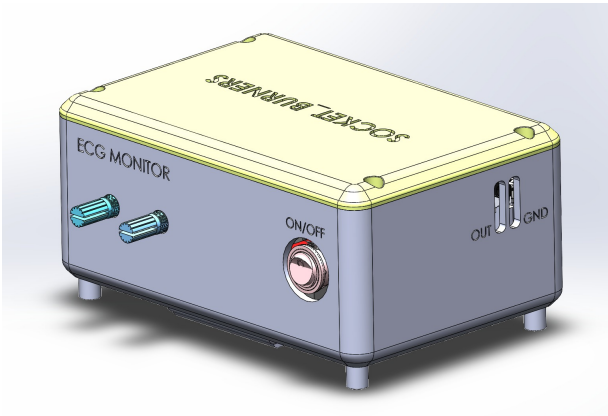

Introduction

PulseView is a portable, battery-powered ECG viewing device designed for clear, real-time visualization of heart electrical activity. It uses a simple three-electrode configuration (RA, LA, RL) and a fully analog signal-conditioning chain to capture clean, high-fidelity cardiac signals without the need for complex digital processing. The device amplifies and stabilizes the micro-volt ECG signal using an instrumentation amplifier, while a Right-Leg Drive circuit actively suppresses common-mode noise for improved clarity. The signal is then refined through a sequence of analog filters, including low-pass filtering to reduce baseline drift, high-pass filtering to suppress muscle-related high-frequency interference, and a 50 Hz notch filter to eliminate ambient power-line noise. The resulting waveform can be observed directly on an oscilloscope or digitized via an ADC for display. PulseView offers a compact reliable solution for portable ECG monitoring for educational applications.

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1 Device Overview

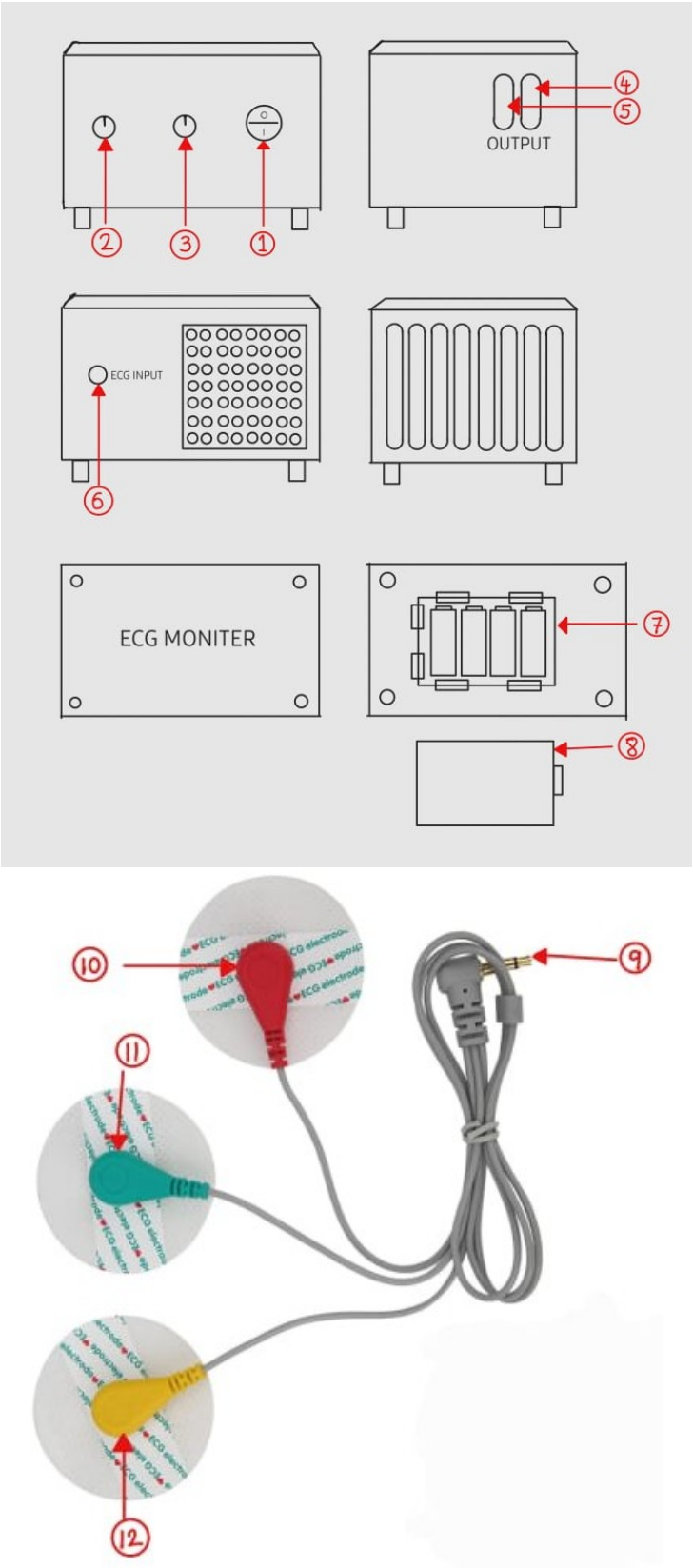

Figure 1: PulseView Device.

Table 1: Device Components

No.	Component Description
1	On/Off switch
2	Instrumentation amplifier gain adjustment knob
3	Notch filter adjustment knob
4	Output signal port
5	Output ground port
6	Auxiliary input for ECG lead
7	Battery compartment
8	Battery cover
9	Auxiliary port of the ECG lead
10	Red lead (Left arm)
11	Green lead (Right arm)
12	Yellow lead (Right leg)

2 Using PulseView Device

2.1 Providing Power

The device is powered by four 18650-type 3.7 V Li-Ion batteries. To insert the batteries, slide open the bottom cover and place the batteries according to the polarity markings on the casing. After closing the cover, toggle the main On/Off switch to power on the device.

2.2 Providing Input

Connect the auxiliary port of the ECG lead to the ECG input [(6) in Figure 1]. Then attach the three electrodes to the patient's chest according to Figure 2.

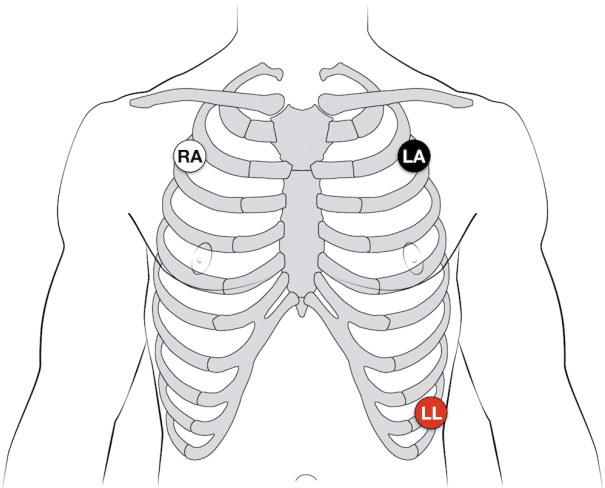

Figure 2: Electrode Placement on the chest

2.3 Adjusting the Signal

The instrumentation amplifier gain adjustment knob [(2) in Figure 1] can be used to control the signal amplitude, while the notch filter adjustment knob [(3) in Figure 1] is used to reduce power-line interference and improve overall signal clarity.

2.4 Maintenance and Care

2.4.1 Cleaning the Device

- Ensure the device is powered off and disconnected from any external equipment before cleaning.
- Wipe the exterior surfaces with a soft, lint-free cloth slightly dampened with water or a mild, non-abrasive detergent.
- Do not use alcohol, solvents, or aerosol cleaners, as they may damage the enclosure or internal components.
- Avoid allowing moisture to enter any openings, connectors, or switches.

2.4.2 Electrode and Cable Care

- Inspect electrode cables regularly for signs of wear, cracking, or loose connectors. Replace immediately if damaged.
- Clean reusable electrodes (if applicable) using a damp cloth. Allow them to dry fully before use.
- Store electrodes with their protective caps or covers to prevent contamination or bending of contact surfaces.
- When disconnecting leads, pull from the connector, not the cable, to prevent internal wire damage.

2.4.3 Battery Care

- Use only the recommended battery type specified for PulseView.
- Remove the battery if the device will not be used for an extended period to prevent leakage or corrosion.
- Avoid exposing the device or battery to excessive heat, direct sunlight, or high humidity.
- If battery performance noticeably decreases, replace it with a new unit of the same specification.

2.4.4 Storage Conditions

- Store the device in a cool, dry place, away from dust and direct sunlight.
- Recommended storage temperature: 10°C–35°C.
- Keep the device away from strong magnetic fields, heavy vibration, and moisture.
- Before long-term storage, clean the device and remove the battery.

2.4.5 Handling and Usage Guidelines

- Avoid dropping, striking, or applying excessive pressure to the device.
- Do not attempt to open or modify the internal circuitry; this may void the warranty and affect performance.
- Keep the device away from liquids and conductive materials.
- Handle the device by its body and connectors, not by attached cables.

3 Troubleshooting

The following outlines common issues encountered during operation of the PulseView ECG device and provides recommended corrective actions. If a problem persists after performing these checks, discontinue use and contact technical support.

3.1 No Signal Displayed

Possible Causes:

- Electrodes not properly attached
- Lead cables disconnected or damaged
- Battery fully depleted
- Loose oscilloscope or ADC connection

Solutions:

- Ensure RA, LA, and RL electrodes are firmly attached to clean, dry skin
- Verify all cables are fully inserted and undamaged
- Replace or recharge the batteries
- Check that the output is correctly connected to the oscilloscope or display module
- Reset the device and restart measurement

3.2 Excessive Noise or Unstable Waveform

Possible Causes:

- Device not tuned properly
- Poor electrode contact or dried gel
- Movement artifacts from patient motion
- Nearby electronic interference
- Damaged or loose cables
- Insufficient grounding or RL electrode not connected

Solutions:

- Adjust the knobs (especially the Q-Factor) to sharpen the signal
- Reapply or reposition electrodes; clean skin before attachment
- Keep the subject relaxed and still during measurement
- Move the device away from AC power sources, motors, or fluorescent lights
- Inspect and replace damaged cables
- Ensure the RL (Right Leg Drive) electrode is properly connected to enable noise cancellation

3.3 Weak or Distorted ECG Waveform

Possible Causes:

- Device not tuned properly
- Incorrect electrode placement
- Baseline drift caused by poor skin contact or movement
- Filters not functioning as expected
- Battery voltage too low for proper amplifier operation

Solutions:

- Tune the device to increase the gain and sharpen the signal
- Confirm electrode placement (RA, LA, RL)
- Improve skin preparation (clean and dry before attaching electrodes)
- Replace the battery and retest

3.4 Device Not Powering On

Possible Causes:

- Battery is dead or improperly inserted
- Power switch fault
- Internal wiring issue or blown fuse (if applicable)

Solutions:

- Replace the battery with a fresh one
- Ensure the battery is inserted with correct polarity
- Toggle the power switch several times to confirm operation
- If the device still does not power on, contact support — do not attempt internal repairs

4 Technical Support Information

If you require assistance with your PulseView device, our technical support team is available to help diagnose issues, provide usage guidance, and arrange repairs when necessary. Please review the troubleshooting section before contacting support, as many common issues can be resolved quickly.

4.1 Customer Support Contact

For questions, general assistance, or technical issues, please contact:

- **Email:** *support@pulseview.com*
- **Phone:** *+94-772455800*
- **Website:** *www.pulseview.com/support*

Support is available Monday–Friday, 9:00–17:00 (local time), excluding public holidays.

4.2 Warranty Information

PulseView is covered by a limited warranty that protects against defects in materials and workmanship under normal use.
Warranty Period: 12 months from the date of purchase.
Coverage Includes:

- Internal electronic components
- Manufacturing defects
- Early component failure

Not Covered:

- Physical damage due to misuse, impact, or improper handling
- Damage caused by unauthorized modifications or repairs
- Normal wear of electrodes, cables, or consumable accessories
- Damage resulting from moisture, excessive heat, or improper storage

Proof of purchase is required for warranty service.

4.3 Repair and Service Procedures

If your device requires service:

1. Contact technical support to describe the issue.
2. Complete basic diagnostics as instructed by support staff.
3. If repair is required, you will receive instructions for shipping or delivering the device to an authorized service center.
4. After inspection, you will be notified of warranty approval or estimated repair costs and turnaround time.
5. Devices approved under warranty will be repaired or replaced at no cost unless conditions invalidate coverage.

Do not attempt to open or repair the device yourself, as this may cause further damage and void the warranty.

4.4 Authorized Service Centers

Service and repairs are handled only by certified facilities to ensure safety and performance standards.

- **Primary Service Center:**
  PulseView Technical Service Unit
  Metro Street,
  Colombo

Additional service locations may be listed on our website.

4.5 Returns & Replacement Policy

If the device was purchased directly from our company or an authorized distributor:

- Returns are accepted within 30 days of purchase if the product is unused and in original condition.
- A replacement unit may be issued if the device is found defective upon arrival.
- Shipping instructions and return authorization will be provided by support.

5 Regulatory Information

PulseView is designed as a low-risk electrocardiography viewing device. The following regulatory information is provided to help users understand the device's intended use and applicable requirements in regions where medical regulations apply.

5.1 Medical Device Status

PulseView is intended for basic ECG monitoring and educational use. It is not intended for clinical diagnosis, patient treatment, or life-support applications unless it has been specifically cleared or approved by the appropriate regulatory authority.
Depending on the region, the device may fall under:

- **United States (FDA):** Classified as a Class I medical device (exempt from premarket notification) when used for general ECG monitoring. Diagnostic use may require FDA 510(k) clearance.
- **European Union (EU MDR):** Typically considered a Class I device for non-diagnostic monitoring. Diagnostic use requires conformity assessment and CE marking.
- **Other Regions:** Local regulatory agencies may classify the device differently based on intended use.

Users must comply with local regulations regarding operation and distribution.

5.2 Intended Use Notice

- The device is intended for recording and viewing ECG waveforms from surface electrodes.
- It is not designed to replace professional medical diagnostic equipment.
- Interpretation of ECG signals should be performed only by trained personnel where diagnostic use is permitted.

5.3 Regulatory Warnings

- Do not modify the hardware or circuitry. Alterations may invalidate regulatory compliance.
- Use only approved accessories and components.
- When used in clinical or educational environments, follow all required safety and electrical standards for the facility.

5.4 Certification Documentation

A formal Regulatory Compliance Statement or Declaration of Conformity may be provided upon request. Users requiring documentation or regulatory assistance should contact technical support.